Rewarding Learning

ADVANCED

General Certificate of Education

2018

Digital Technology

Assessment Unit A2 1

assessing

Information Systems

[CODE]

SPECIMEN

MARK SCHEME

Version 2: 6 March 2017

AVAILABLE MARKS

1 (a) Server

Manages shared network resources [1] such as data/files/peripherals [1]. It processes requests from other nodes [1]

Any **two** points [2]

Wireless access point

To connect wireless devices [1] to a wired network [1] using Wi-Fi or Bluetooth [1]

Any **two** points [2]

MAC address

Assigns an unique address [1] to each node on the network [1] which identifies the network interface card [1]

Any **two** points [2]

(b) In a peer to peer network, all nodes have equal status [1]
Nodes share their resources with one another [1]
In a server based network, one node controls the network and its resources [1]
Other nodes request resources from the server [1]
Remote software upgrades are possible with client server [1]
Peer to peer upgrades managed independently on each node [1]

Any **four** points [4]

(c) The nodes are connected in a loop [1]
Each node may be connected directly to the ring or by a spur [1]
Data passes from node to node [1] until it reaches its intended recipient [1]
Data travels in one direction [1]

Any **three** points [3]

(d) Security

The star topology is more secure [1] because the server sends data directly to its intended recipient [1]. In the ring topology, data is sent from node to node [1] until it reaches intended recipient [1]

Any **three** points [3]

Reliability

The star topology is more reliable [1] because if a link fails, only one node is affected [1]. In the ring topology, if a link fails the loop fails [1]
If the central server fails in a Star network then whole network goes down [1]

Any **three** points [3]

(e) This operates at hardware level [1]
It transmits and receives streams of bits [1]
It specifies voltage levels [1], specifies timings [1]

Any **three** points [3]

(f) **Bandwidth**
Fibre optic cable provides more bandwidth than metal cables [1] because a single cable consists of multiple strands [1] each strand capable of carrying multiple data signals [1]. A metal cable only has a capacity equal to one of these strands [1]

Any **three** points [3]

Security
Fibre optic cable is more secure [1] as it is difficult to intercept its signal without destroying it [1]. The signal in a metal cable can be detected electromagnetically [1]. Physical casing is more secure [1]. Metal cable is more susceptible to interference due to copper medium [1]; where as fibre optic uses light medium [1]

Any **three** points [3]

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2 **(a)** Data is stored in tables [1] consisting of records [1]
Each record is uniquely identified by a primary key [1]
Tables are linked via primary keys [1] and foreign keys [1]

Any **four** points [4]

(b) **Entity**
An object/subject about which information is stored [1] such as Member/Book/Loan [1] [2]

Attribute
Each data item within an entity [1] such as Title in Book [1] [2]

Primary key
An attribute which uniquely identifies an entity occurrence/record [1] such as BookID [1] [2]

Composite key
A composite key consists of two or more key fields each of which may be a primary key in another table/which together uniquely identify an entity occurrence [1] such as MemberID+BookID [1] [2]

(c)



[1] + [1] for two correct relationships
[1] for all three entities
[1] for proper ERD symbols [4]

			AVAILABLE MARKS
	(d)	Data redundancy is minimised [1] which minimises storage requirements [1] [2] Data inconsistency is minimised [1] which improves data integrity [1] [2]	
	(e)	INSERT INTO Loan [1] VALUES (43216,21345, 24/08/2016, 31/08/2016, False) [1] + [1] [3]	
	(f)	SELECT Title, Author [1] FROM Book [1] WHERE Category = “Science” [1] [3]	
	(g)	Using QBE, the librarian can enter the search criteria [1] by selecting an option from a list [1] or by entering values in text boxes [1]. This requires less expertise than using SQL whose commands have a complicated syntax [1] [4]	30
3	**(a)**	The mobile phone uses a special radio frequency [1] to contact the nearest mobile phone mast [1] A separate radio frequency is used to transmit the phone conversion [1] Each phone mast controls a cell [1] Adjacent cells overlap to ensure maximum coverage [1] If there is more than one phone mast in range, the mast receiving the strongest signal from the mobile phone is used [1] As the phone user moves out of the mast’s range, a nearby mast will take control of the call [1] A local base station controls a number of phone masts [1] The local base station routes the phone call to its mobile switching centre [1] Any **six** points [6]	
	(b)	**Expert system shell** This assists/automates the creation of the expert system [1] It is an empty or skeletal expert system [1] For a particular application, the relevant rules and facts are added [1] [3] **Fuzzy logic** Deal with uncertainties [1] Statements are assigned probabilities [1] rather than being either true or false [1] [3] **Heuristics** These are rules [1] derived from human experience [1] or intuition [1] [3]	
	(c)	**Indicative content** Candidates may discuss the following: The expert system’s knowledge base contains information and rules about illnesses and symptoms. The patient responds to questions and the expert system provides a diagnosis of the patient’s illness.	

Advantages

The expert system's knowledge can reflect the knowledge of a large number of medical experts
The expert system will produce accurate and consistent diagnoses
Make rational decisions without any emotional overhead
Knowledge of expert staff can be captured to some extent before they move on to new locations or jobs
Can be used as a training aid to increase the expertise of staff
Efficient way of providing answers to queries (reducing wait times) as it does not involve additional help to staff
Natural language interface would make the expert system more human friendly

Disadvantages

The expert system cannot use intuition
Overreliance on expert systems could lead to a loss of human expertise/skills
A lot of effort and cost has to go into making a good expert system
Not as good as having human experts to hand
Most expert systems are menu driven which does not deal very well with ambiguous problems
Does not learn from mistakes unless user feedback and human maintenance is part of its ongoing development

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Level of Response	Marking Critera	Marks
Band 3: Excellent	The candidate describes accurately and coherently how the expert system is used in the hospital. The candidate articulates clearly the main advantages and disadvantages of using an expert system in this context. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]−[9]
Band 2: Good	The candidate describes broadly how the expert system is used in the hospital. The candidate identifies some advantages or disadvantages of using an expert system but may not refer to the hospital. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]−[6]
Band1: Basic	The candidates attempts to explain how the expert system might be used in the hospital. The candidate identifies an advantage or disadvantage of using an expert system but does not refer to the hospital. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]−[3]
	The response is not worthy of credit	[0]

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			AVAILABLE MARKS
4	**(a)**	Data mining involves the automatic analysis [1] sorting of large data sets in a data warehouse [1]. Information is combined from a number of sources [1]. The aim is to identify trends [1] and correlations [1] Any **four** points [4]	
	(b)	Supermarkets mine point-of-sale data [1] and data collected from loyalty cards [1] to create accurate customer profiles [1] and to improve their targeting of products and services [1] They can identify the products whose sales are most susceptible to promotional events [1] They can get a clear picture of demand for product ranges and ensure stores are appropriately stocked [1] They also mine socio-economic data [1] and data about competitors [1] to help them devise marketing strategies [1] Any **seven** points [7]	11
5	**(a)**	Personal data must be: 1 Processed fairly and lawfully. 2 Obtained for specified and lawful purposes. 3 Adequate, relevant and not excessive. 4 Accurate and up to date. 5 Not kept any longer than necessary. 6 Processed in accordance with the “data subject’s” (the individual’s) rights. 7 Securely kept. 8 Not transferred to any other country without adequate protection in situ. Any **four** principles [4]	
	(b)	Right of access [1] The data subject has a right to be supplied with the personal data held about them [1] Right of correction [1] The data subject may force a data controller to correct any mistakes in the data held about them [1] Right to prevent distress [1] The data subject may prevent the use of information if it would be likely to cause them distress [1] Right to prevent direct marketing [1] The data subject may stop their data being used for junk mail or cold calling [1] Right to prevent automatic decisions [1] The data subject may specify that they do not want a data user to	

make automated decisions about them Example: where, through points scoring, a computer decides on a loan application [1]

Right of complaint [1]
The data subject can ask for the use of their personal data to be reviewed by the Information Commissioner who can enforce a ruling [1]

[2] for each of any **two** rights [4]

(c) Indicative content

Candidates may discuss the following:

In general, legislation lags behind developments in Digital Technology and ethics fill the gap. The increasing and expanding use of Digital Technology for gathering, storing, manipulating, and communicating data
…has revolutionised the use and spread of information and has created ethical dilemmas.

Personal data can be used for purposes quite different from its intended use
…because the legislation governing who can use the data and for what purpose may not be clear or restrictive enough
…to protect the intentions and interests of the subjects.

Companies may obtain data legitimately from a number of third-party sources but combine this data in ways of which the data subject has no understanding. Companies may obtain medical, financial and criminal records even though they have no relevance to their business.

Users of data must recognise changes in rights and responsibilities
…and adapt their standards of conduct which date from before the advent of computers.

Level of Response	Marking Criteria	Marks
Band 3: Excellent	The candidate articulates clearly a range of ethical considerations relating to the use of personal information with specific reference to the broader consequences for the companies using the data. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]−[9]
Band 2: Good	The candidate describes some ethical considerations relating to the use of personal information by companies. The candidate makes some reference to the consequences for organisations. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]−[6]
Band 1: Basic	The candidate attempts to describe some issues but fails to distinguish between legal and ethical considerations. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]−[3]
	The response is not worthy of credit	[0]

AVAILABLE MARKS

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6 (a) Virtualisation

The creation of a virtual version of a device or resource [1] where users (devices, applications or human users) can interact with the virtual resource as if it were a real single resource [1]
Storage virtualisation [1] is the amalgamation of multiple network storage devices into a single storage unit [1]
Server virtualisation [1] is the partitioning of a physical server into smaller virtual servers [1]
Operating system virtualisation [1] where multiple operating systems can be used on a single server [1]
Network virtualisation [1] in which all servers and services in the network are treated as a single pool of resources that can be accessed without regard for its physical components [1]
Application virtualisation [1] where each application adapts its set of configurations on-demand [1]

Any **six** points [6]

(b) Indicative content

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Candidates may discuss the following:

The third party will provide the payroll system via a URL and it will maintain the organisation's payroll database.
The organisation will use this URL to run its payroll instead of running the payroll system on its own network and maintaining the payroll database themselves.
The organisation does not need to implement its own security specifically for their payroll operations.
Security is implemented by the third party.
This removes control over security from the organisation as they have to rely on the third party's security.

Level of Response	Marking Criteria	Marks
Band 3: Excellent	The candidate articulates clearly how the organisation will use a third party in this case. The candidate describes clearly and accurately the security considerations, including the key advantages and disadvantages. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]−[9]
Band 2: Good	The candidate describes how the organisation will use a third party in this case. The candidate addresses the issue of security, and identifies some advantages or disadvantages. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]−[6]
Band 1: Basic	The candidate attempts to explain how the organisation will use a third party in this case. The candidate does not address the issue of security, but identifies an advantage or disadvantage. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]−[3]
The response is not worthy of credit		[0]

(c) Indicative content

Candidates may discuss the following:

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Instead of storing data on the personal computer's external memory
…such as the hard disk/DVD/flash memory
…data is uploaded to the internet
…where it is stored on a remote server
... unlimited storage capacity
...host is responsible for all updates
...host is reponsible for all backups
This has reduced the need for high capacity external memory on personal computers
This has not reduced the need for internal memory
Device independence
Increased data reliability

Level of Response	Marking Criteria	Marks
Band 3: Excellent	The candidate articulates clearly how a personal computer user can make use of cloud computing. The candidate explains in detail the main consequences on the resource requirements of the personal computer. The candidate uses the appropriate Digital Technology vocabulary accurately. Presentation, spelling, punctuation and grammar are of a high standard.	[7]−[9]
Band 2: Good	The candidate describes how a personal computer user can make use of cloud computing. The candidate describes some consequences on the resource requirements of the personal computer. The candidate uses some relevant Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make the response clear.	[4]−[6]
Band 1: Basic	The candidate refers to a personal computer user using cloud computing. The candidate makes some reference to the use of resources on the personal computer. The candidate makes limited use of Digital Technology vocabulary. Presentation, spelling, punctuation and grammar are such that the intended meaning is not clear.	[1]−[3]
	The response is not worthy of credit	[0]

AVAILABLE MARKS

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		AVAILABLE MARKS
7 (a)	**Parity bits** A bit is added to a group of bits to make its sum odd (or even) [1] Example: the 8th bit in a byte will be the parity bit [1] When a byte is transmitted its parity is checked [1] If the parity of the byte has changed, at least one bit in the byte has flipped [1] but an even number of flips will not be detected [1] Any **four** points [4]	
	CRC This is a special checksum [1] applied to a block of data [1]. An algorithm or polynomial [1] is applied to the block and the checksum appended to the block [1]. The recipient applies the same algorithm to the block and compares its result with the result already appended [1]. If they do not agree, the data has not been received successfully [1] Any **four** points [4]	
(b)	**SSL** This is secure protocol [1] for financial transactions over the Internet [1]. This enables an encrypted link between two computers on a network [1]. It enables a browser to identify the web server with which it is communicating [1]. It uses an SSL Certificate [1] Any **four** points [4]	
	Digital certificate A digital code is generated [1] authenticated [1] by public key encryption [1] and attached to an electronically transmitted document [1] to verify its contents [1] and the sender's identity [1] Any **four** points [4]	16
	Total	**150**